

ANSWER KEY — Parent / Teacher Copy**Chemistry — Unit 5: Molecular Shape**

Quiz [A] — Bonding Sites, Lone Pairs, Shape Names, and 3-D Drawing

Section 1 — Identify Bonding Sites and Lone Pairs (5.1.1)**1. BF_3**

(4 pts — 2 pts each)

- a. Bonding sites on B: **3**
- b. Lone pairs on B: **0**

Why:

Boron has only 3 valence electrons and forms 3 single bonds (one to each F). There are no electrons left over for a lone pair. Every electron boron has is used in a bond. This is one reason BF_3 is unusual—B has an “incomplete octet.”

2. H_2O

(6 pts — 2 pts each)

- a. Bonding sites on O: **2**
- b. Lone pairs on O: **2**
- c. Total electron pairs around O: **4** (2 bonds + 2 lone pairs)

Why:

Oxygen has 6 valence electrons. It uses 2 to bond with the two H atoms (1 electron per bond shared with H). That leaves 4 electrons = 2 lone pairs. Total around O = 4 electron pairs, which gives it a *tetrahedral domain geometry* even though the shape is bent.

3. SO_2

(6 pts — 2 pts each)

- a. Bonding sites on S: **2** (*double bond to one O = 1 site; single bond to other O = 1 site*)
- b. Lone pairs on S: **1**
- c. Difference from SO_3 : **SO_3 has 3 bonding sites and 0 lone pairs (trigonal planar, 120°); SO_2 has 2 bonding sites and 1 lone pair (trigonal planar bent, $<120^\circ$). The lone pair on S in SO_2 changes both the shape name and the bond angle.**

Section 2 — Name Molecular Shapes (5.1.2)**4. Fill in the blanks**

(4 pts — 1 pt each blank)

- a. 2 bonding sites, 0 lone pairs → Shape: **Linear** Bond angle: **180°**
- b. 4 bonding sites, 0 lone pairs → Shape: **Tetrahedral** Bond angle: **109.5°**

Memory tip for Marcus:

Linear = 2 atoms pulled as far apart as possible = straight line. Tetrahedral = 4 bonds pushing equally in 3D space = 109.5° is the “perfect” spread of 4 equal forces in 3D. Think of CH_4 as a pyramid with 3 base H’s and 1 top H; all angles equal.

5. 3 bonding sites, 1 lone pair

(6 pts — 2 pts each)

- Shape name: **Trigonal Pyramidal**
- Approximate bond angle: **107°**
- Why smaller than 120° : **The lone pair creates a 4th electron domain around the central atom. With 4 total electron domains (3 bonds + 1 lone pair), the domain geometry is *tetrahedral* and the expected angle is 109.5° . But the lone pair repels more strongly than a bonding pair, compressing the bond angle from the tetrahedral ideal of 109.5° down to $\sim 107^\circ$. It is smaller than 120° because the molecule has 4 electron domains, not 3.**

Key concept:

Lone pairs are “fatter” than bonding pairs because they’re only attracted to one nucleus (the central atom) instead of two. They spread out more and squeeze the bonds. NH_3 is the classic example: domain geometry is tetrahedral (4 pairs), so the base angle is 109.5° —not 120° . The lone pair then squeezes that down to $\sim 107^\circ$. The shape name is trigonal pyramidal because we count only atoms, not lone pairs.

6. Complete the table

(6 pts — 2 pts per row)

Molecule	# Bonding Sites	# Lone Pairs	Shape Name
CH ₄	4	0	Tetrahedral
NH ₃	3	1	Trigonal Pyramidal
H ₂ O	2	2	Tetrahedral Bent (Bent)

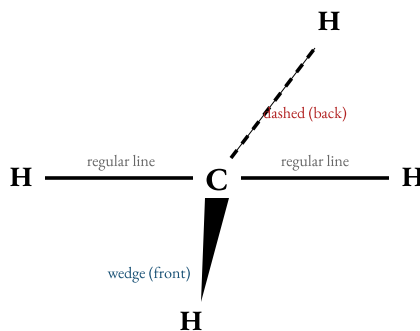
Pattern to notice:

CH₄, NH₃, and H₂O all have 4 total electron pairs (domain geometry = tetrahedral). But as you replace bonding pairs with lone pairs, the shape changes and the bond angle gets smaller: 109.5° (CH₄) → 107° (NH₃) → 104.5° (H₂O). This is a key pattern Kai should know cold.

Section 3 — Draw 3-D Structures (5.1.3)

7. CH₄ — Tetrahedral 3-D Drawing

(8 pts)

**Scoring rubric (8 pts):**

1 pt — Wedge bond present and drawn as a filled/shaded triangle pointing toward viewer

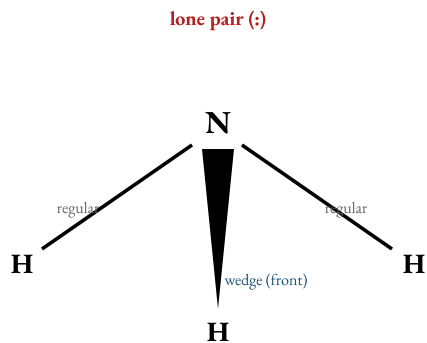
1 pt — Dashed bond present and drawn as a dashed line pointing away from viewer

2 pts — Two regular lines drawn in the plane of the paper (left and right)

4 pts — H atom correctly placed at the end of each of the 4 bonds (1 pt each)

*Accept any orientation as long as bond types are correctly labeled and central atom is C.*8. NH₃ — Trigonal Pyramidal 3-D Drawing

(10 pts)



Sample explanation (full credit):

“ NH_3 is trigonal pyramidal, not tetrahedral, because one of the four electron pairs around N is a lone pair, not a bond. Molecular shape is named based on where the *atoms* are, not the lone pairs—so with only 3 atoms bonded to N, the shape is a triangular pyramid with N at the top, not a tetrahedron.”

Scoring rubric (10 pts):

- 3 pts — Drawing shows correct shape (N at top, 3 H at base, pyramidal)
- 3 pts — Bond types correctly labeled (at least 1 wedge distinguishable from regular bonds)
- 1 pt — Lone pair shown/labeled above N
- 3 pts — Explanation mentions: (1) lone pair present, (2) shape based on atom positions only, (3) correct shape name

Total: _____ / 50